

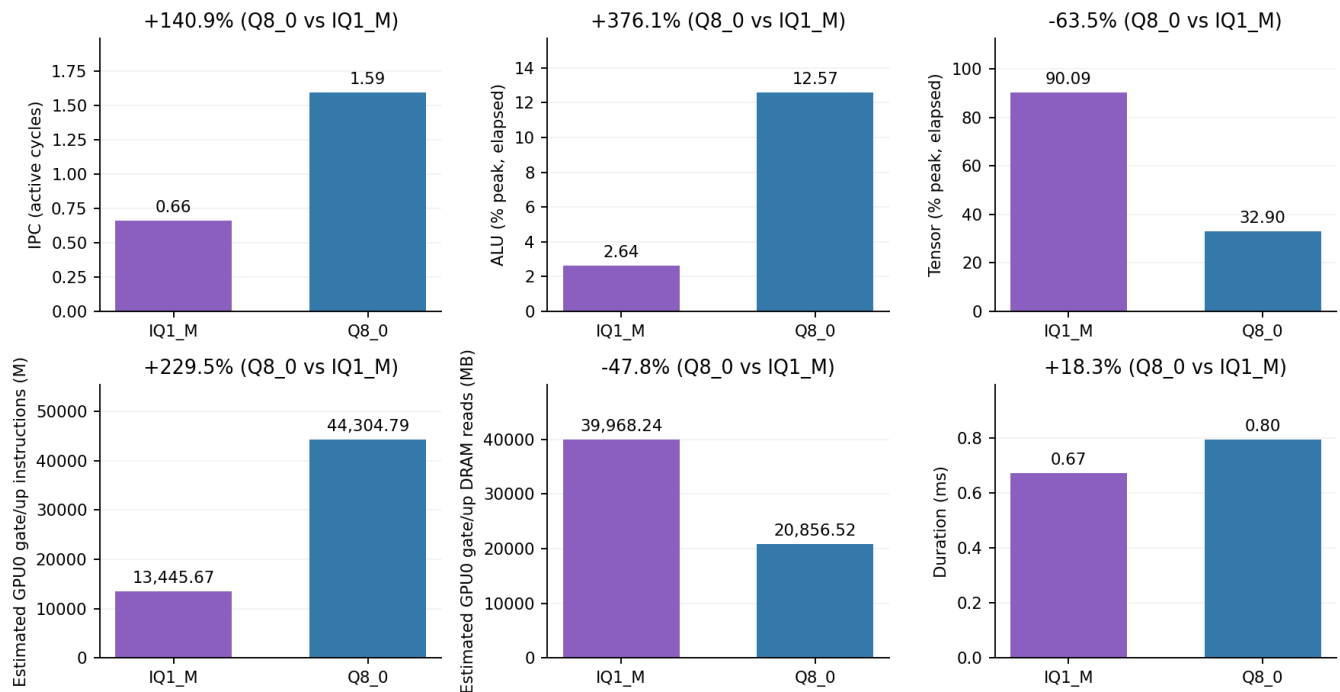
Runtime bottlenecks of IQ1_M and Q8_0 quantization

Llama 3.3 70B; two GPUs: NVIDIA RTX PRO 6000 Blackwell Server Edition; C8; 2,048 input and 512 output tokens; FP16 KV.

Reduced image scope: eight full-section captures sample five gate/up launches per format, phase and GPU. Values labelled estimated scale each per-launch median across the gate and up operations in the transformer layers assigned to that GPU. The scalar roofline reuses these captures; no separate operator inventory was collected.

GPU0 prefill: Q8_0 kernel duration changes +18.3%, executed instructions +229.5%, and DRAM reads -47.8% relative to IQ1_M. Duration is observed per launch; labelled instruction and DRAM totals are layer-placement estimates.

Prefill gate/up - GPU0 - M=28672, N=512, K=8192; estimated totals span 41 layers



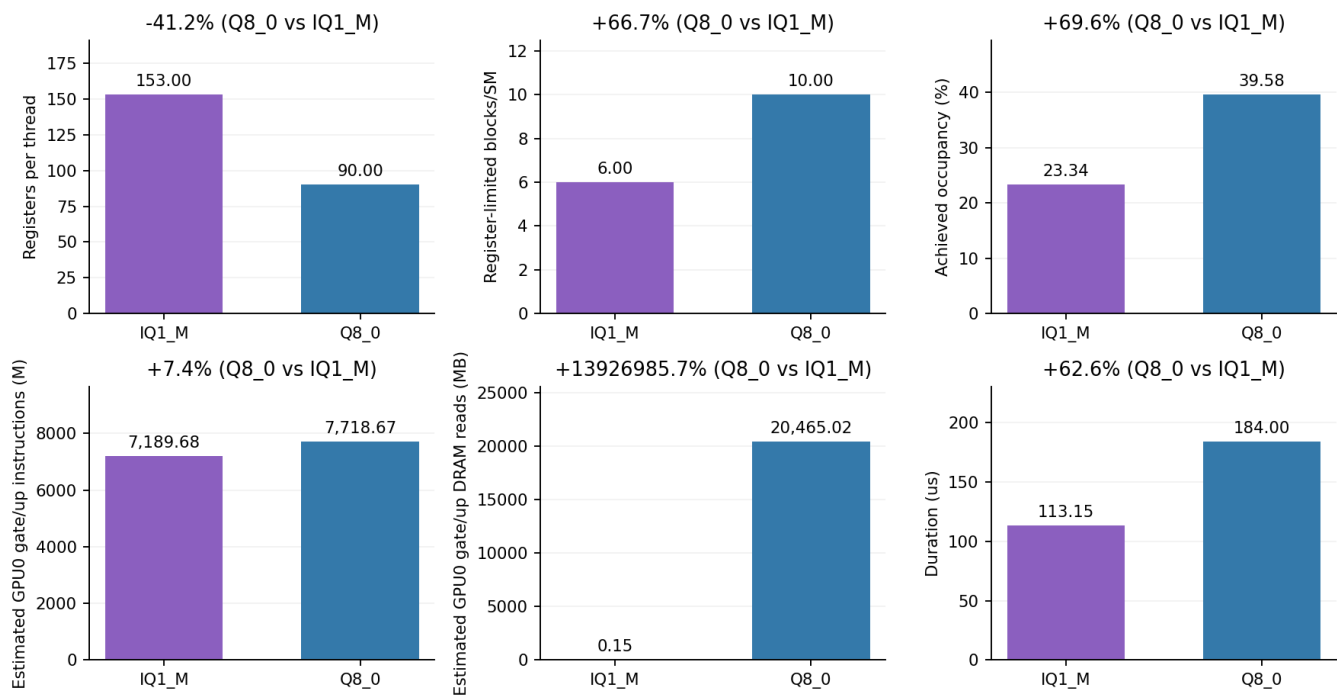
Five sampled gate/up launches per format/phase/GPU; medians within a matched matrix geometry and a single kernel configuration. Pipeline percentages describe different resources and must not be added. Register pressure and occupancy do not by themselves isolate a latency cause.

Actual kernel names and matrix geometry are retained in report-data.json; conclusions from another quantization pair are not transferred to this shard.

Decode gate/up measurements

GPU0 decode: Q8_0 kernel duration changes +62.6%, executed instructions +7.4%, and DRAM reads +13926985.7% relative to IQ1_M. Duration is observed per launch; labelled instruction and DRAM totals are layer-placement estimates.

Decode gate/up - GPU0 - M=28672, N=8, K=8192; estimated totals span 41 layers

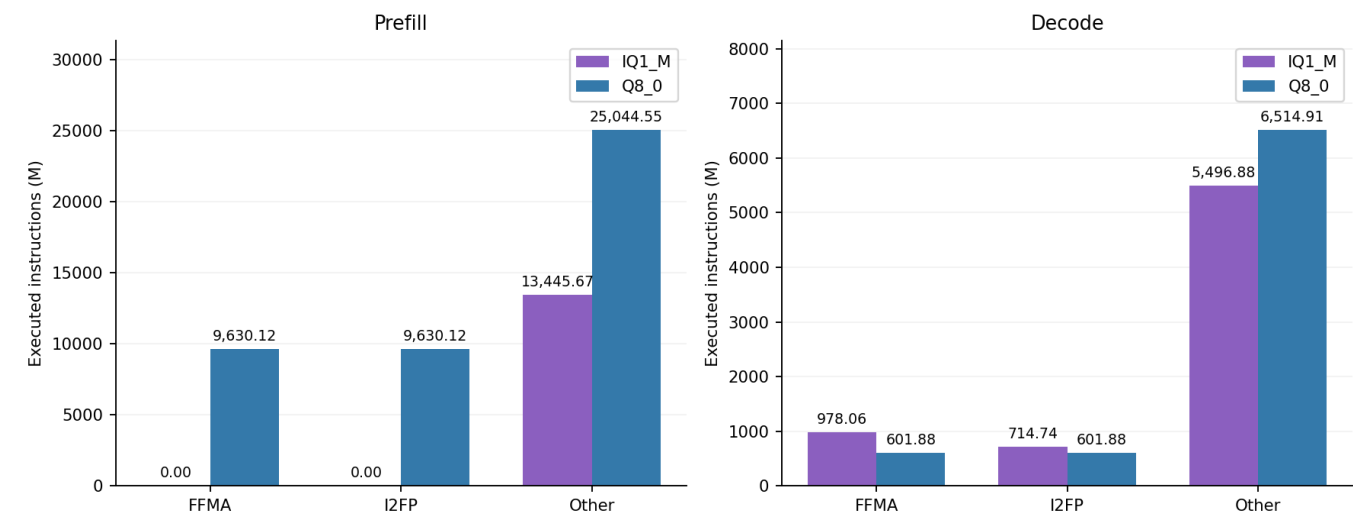


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Executed instruction mix

Estimated gate/up instruction mix across GPU0 layers

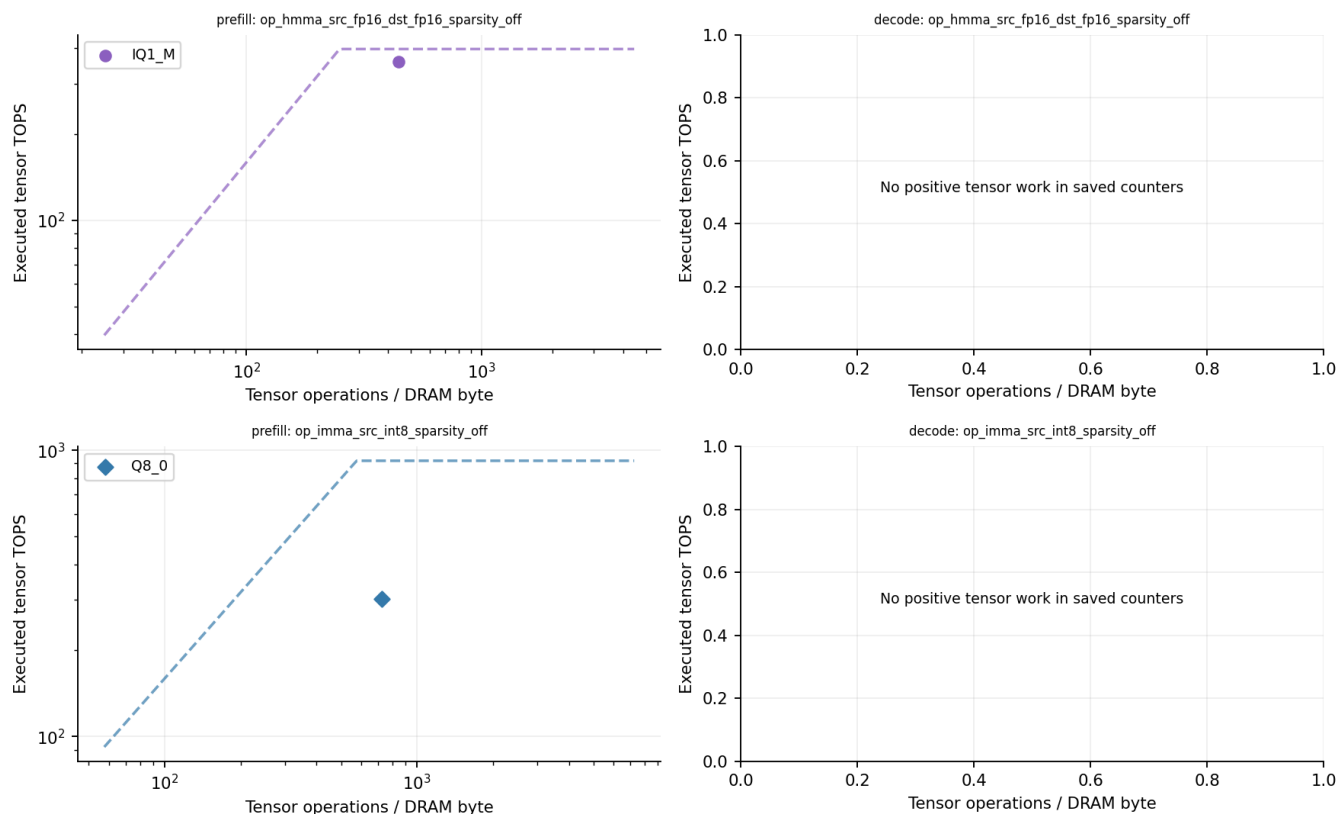


Phase / opcode	IQ1_M (M)	Q8_0 (M)	Change
prefill / FFMA	0.00	9,630.12	undefined
prefill / I2FP	0.00	9,630.12	undefined
prefill / Other	13,445.67	25,044.55	+86.3%
decode / FFMA	978.06	601.88	-38.5%
decode / I2FP	714.74	601.88	-15.8%
decode / Other	5,496.88	6,514.91	+18.5%

FFMA and I2FP counts are read from complete SASS opcode inventories and checked against total instructions per launch. An absent opcode is zero only after that reconciliation. Other includes every remaining opcode. The displayed totals multiply per-launch medians by two gate/up operations per placed transformer layer; they estimate one matched matrix batch. Instruction-count differences do not establish separate runtime costs.

Tensor Core arithmetic paths

Tensor Core roofline - GPU0 - actual arithmetic paths and same-capture peaks

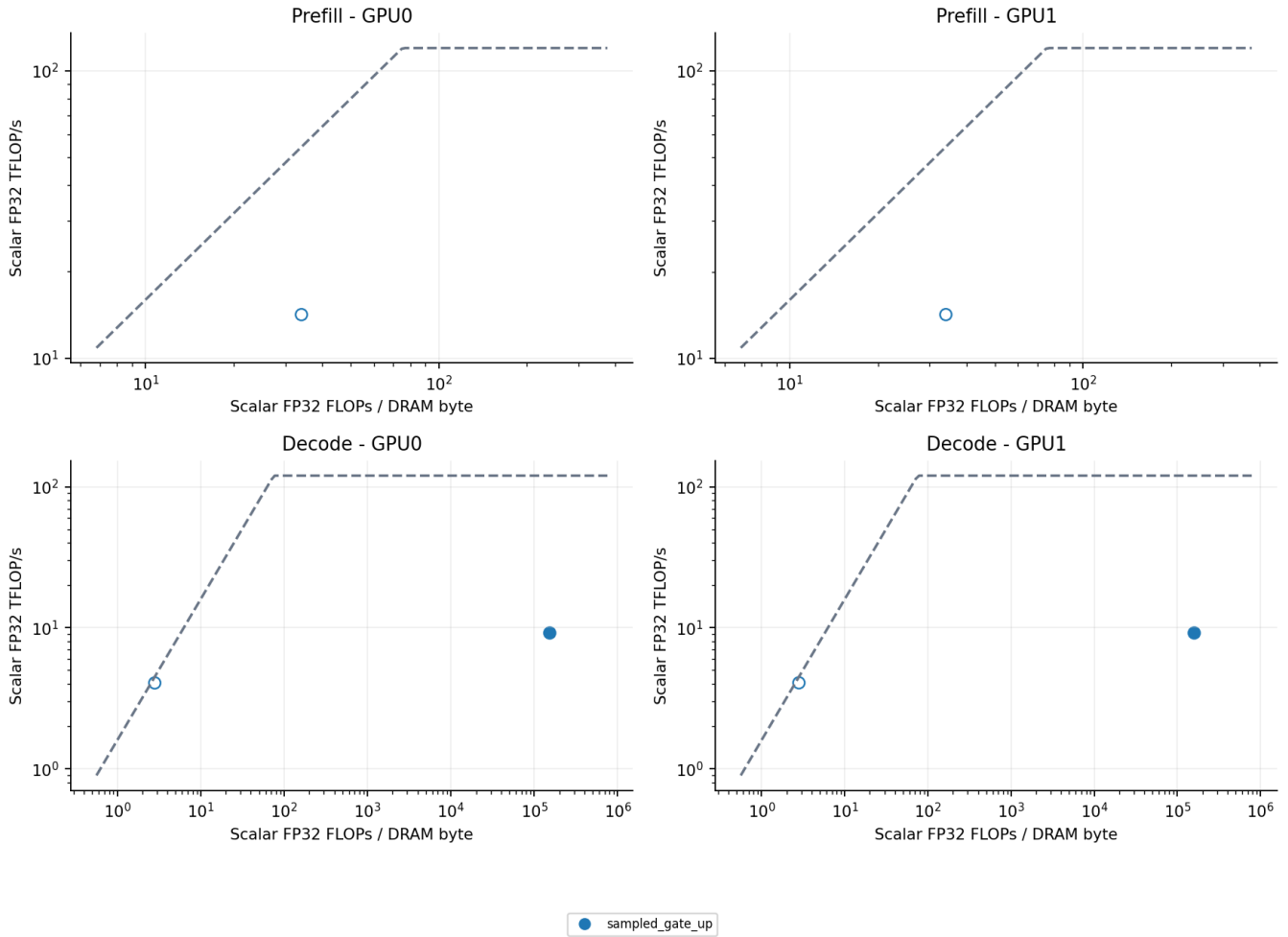


Work and matching peak counters are selected by their actual exported arithmetic path, including Blackwell op_imma/op_hmma paths. Each coordinate is calculated per launch before aggregation. Dashed ceilings use same-capture sustained peaks and observed clocks; Quantized weight storage bits are not a compute precision. Zero-work paths remain in tensor-roofline-samples.csv and have no logarithmic point.

The plot covers tensor work in the selected kernels, not whole-serving performance or a unique bottleneck classification.

Sampled gate/up scalar FP32 rooflines

Sampled gate/up scalar FP32 rooflines - IQ1_M filled, Q8_0 hollow



Per-GPU rated ceilings: 120 TFLOP/s and 1597 GB/s. $F = FADD + FMUL + 2*FFMA$ from predicated-on thread counts; $B = \text{DRAM reads} + \text{writes}$; $x = F/B$; $y = F/\text{duration}$. FP16, tensor, integer and special-function work are excluded.

The same eight full-section captures provide 8 gate/up kernel/configuration groups; five matching launches contribute to each group. 2 groups have zero scalar FP32 work and are retained in CSV without invented plot positions. This page describes sampled gate/up kernels only. Normalization, attention, activations, KV updates, the output head and other helpers were not captured.

Both GPUs and reproducibility

Phase	GPU	IQ1_M time (us)	Q8_0 time (us)	Change	Samples/format
prefill	0	672.192	795.520	+18.3%	5
prefill	1	628.160	794.336	+26.5%	5
decode	0	113.152	184.000	+62.6%	5
decode	1	112.096	183.680	+63.9%	5

Each capture uses eight discarded warmup requests and eight profiled requests. Both server GPUs remain visible; only one GPU's counters are selected per capture. NVTX labels establish actual phase, tensor role and matrix dimensions. CUDA graphs are disabled for attribution. Cache and clock controls are none. Profiler replay timings are separate from the unprofiled serving measurements.

Each setting has one capture, not independent repetitions. GPU agreement does not prove universality across workloads. The resulting report describes the saved counters; it does not copy the previous GPU's performance conclusions.

Pinned llama.cpp: 3f5e94d7c2ab2267fe39852051777fe30c1f49ef. Source hashes: evidence.json. Complete launch metrics: hardware-metrics.csv. Opcode counts: instruction-instances.csv. Tensor coordinates: tensor-roofline-samples.csv. Scalar coordinates: operator-roofline-data.csv. Layer projection inputs: gate-up-layer-extrapolation.csv. Rated scalar roofline source: <https://www.nvidia.com/en-us/data-center/rtx-pro-6000-blackwell-server-edition/>